Secure Multi-Satellite Collaborations With ISAC

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***Abstract*—Low Earth Orbit (LEO) satellite systems with sensing capabilities are widely regarded as promoting reliable and efficient communication services globally. This paper proposes a Multi-Satellite Collaborative Security System with Integrated Sensing and Communication (ISAC-MSC). Considering the potential benefits of LEO satellites and ISAC, we exploit sensing performance in the MSC system by jointly optimizing LEO satellite assignments, communication Beamforming (BF) vectors, and sensing BF vectors. Specifically, we design improved Continuous Particle Swarm (CP) optimization and Discrete Particle Swarm (DP) optimization algorithms to maximize the target sensing Signal-to-Noise Ratio (SNR) for LEO satellite assignments. Additionally, with respect to the BF vector optimization, we develop Power Approximation (PA) optimization algorithm, Inner Approximation (IA) optimization algorithm, and Joint Sensing and Communication BF (JSC-BF) optimization algorithm. Multiple algorithms are tightly integrated and alternately iterated. Numerical results show that: 1) the JSC-BF algorithms outperform the PA and IA algorithms in terms of sensing performance and communication secrecy rate; 2) compared to the single satellite case, the ISAC-MSC system performance approximately linear growth, and has strong extensibility; 3) with imperfect MSC synchronization case, the communication secrecy rate appears inflection point and stabilization, but the JSC-BF algorithms still have excellent performance.**

***Index Terms*—Integrated sensing and communication, multi-satellite collaboration, resource allocation, security communications.**

I. Introduction

WITH the expansion of commercial space activities, Multi-Satellite Collaboration (MSC) has become essential for providing multi-layered coverage over target regions for the same mission. However, the open nature of the space environment makes satellite links inherently vulnerable to eavesdropping, which threatens both information security and communication integrity [1], [2], [3].

To counter such threats, satellites require the ability to sense their surroundings and precisely locate potential adversaries for various security applications [4]. Integrated Sensing and Communication (ISAC) provides the technological foundation to meet this demand. The necessity of combining ISAC with MSC becomes apparent, as fusing sensor data from multiple satellites in different orbits significantly boosts the speed and precision of threat localization [5]. Consequently, the core challenge lies in engineering a secure and reliable ISAC-MSC network, which is the cornerstone for maintaining a robust, efficient and globally interconnected satellite communication infrastructure [6].

In recent years, the number of satellites in orbit has been rising to nearly 15000, and the goal of gradually forming a sky-net is being achieved [7]. Due to Low Earth Orbit (LEO) satellites being closer to the ground, usually between 200 and 2000 kilometers, they are suitable for applications that require low latency and high reliability. In the meantime, LEO satellites account for 90 percent of the total number of satellites [8], which is also recognized as a prerequisite for the realization of MSC.

A. Recent Progress in Secure ISAC-MSC

The synchronization of signal transmission and data sharing is a cornerstone of effective MSC systems. To this end, initial synchronization can be efficiently achieved by reducing the time-frequency search space through incoherent weak signal acquisition [9]. For more precise carrier and phase estimation, particularly within secure systems, an iterative code-aided estimation algorithm was proposed in [10]. Furthermore, to address practical challenges like timing variations in large-scale systems, a delay-error-robust algorithm was validated in [11]. Complementing these efforts in synchronization, other work has focused on optimizing data exchange. The authors of [12] addressed this by using an iterative load algorithm to maximize inter-satellite transmission rates, thereby facilitating enhanced data sharing and collaborative scheduling.

Due to its wider spatial scope and open electromagnetic environment, the MSC system faces interference and security threats from multiple sources [13], [14]. Unlike terrestrial secure communications, eavesdroppers exist in more diverse forms, including drones, eavesdropping satellites, and terrestrial terminals [15], [16]. In addition, eavesdroppers may intercept and possibly tamper with downlink signals by masquerading as legitimate satellites or receiving stations [17].

In satellite-ground secure communications, the acquisition of prior information is of paramount importance, serving as

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the key to ensuring communication security and enabling the early identification of risks. The authors of [18] conducted research on this topic, achieving precise estimation of the location of eavesdroppers by assessing the probability of an internal attacker inferring targets' position from received signals. The failure of transmitters to obtain accurate Channel State Information (CSI) is another major challenge in satellite-terrestrial security networks. The satellites improved security communication with imperfect CSI (ipCSI) between satellites and relay nodes by deploying a friendly jammer, where the amplification and forward-based relaying scheme was validated in [19]. Overall, obtaining the status information of the eavesdropper is a prerequisite for enhancing secrecy level.

MSC communication system is vulnerable to attacks, but empowerment through ISAC transforms them into secure systems that can proactively adapt to complex environments. Modern remote sensing satellites are capable of integrating a wide range of sensors, enabling simultaneous ground observation and data acquisition to satisfy the demands for different mission types [20]. In parallel, ISAC was introduced to LEO satellites as a natural development based on the necessity of LEO satellite sensing capabilities [21].

In the downlink ISAC system, the transmitter provided service to users by choosing to steer the beam towards targets within a multi-static sensing framework, where sensing beams improve the detecting performance at lower Radar Cross Sectional (RCS) variance [22]. The ISAC-based LEO satellite system utilizing Beamforming (BF) technology can operate both wireless communication and target sensing with high satisfaction [23]. Moreover, the transmission of quantized dual-function signals can be achieved in LEO satellite systems. With jointly designed ISAC beams, it becomes feasible to implement target detection and user service provision simultaneously for the same satellite [24]. It is crucial that ISAC provides LEO satellites with an active defense mechanism to effectively sense and respond to the threat of eavesdroppers, which greatly enhances security for satellite communications [4].

B. Motivations

The above studies provided the theoretical foundation for the security analysis of the ISAC-MSC system. Currently, threat sources with multi-domain spatial distributions pose severe and complex security challenges to satellite communications. The development of advanced systems equipped with active defense, real-time situational awareness, and eavesdropping threat response capabilities has thus become an urgent need in the field of satellite communications. ISAC technology offers a viable solution for this scenario, enabling satellites to actively sense the target environment during communication operations. However, when applied to a single satellite, ISAC technology encounters insurmountable performance bottlenecks [25]. A single LEO satellite, constrained by its payload limitations and a single observation perspective, exhibits sensing performance that is far insufficient for reliable monitoring of wide-area dynamic threats. This limitation indicates that relying solely on single-satellite ISAC technology cannot meet security requirements, resulting in a critical technological gap in the current landscape.

This paper proposes an ISAC-MSC security system that leverages the sensing and communication resources of multiple satellites to aggregate observational data, thereby overcoming the performance limitations of individual satellites and achieving broader coverage and more precise target sensing. The core issue to be addressed thus emerges, namely maximizing the collaborative detection performance of the ISAC-MSC system through optimized resource allocation, thereby directly enhancing its secure transmission capabilities. Meanwhile, we are the first to consider maximizing target sensing Signal-to-Noise Ratio (SNR) in the ISAC-MSC system. In this paper, we formulate the target sensing SNR maximization by jointly optimizing LEO satellite assignments, hybrid BF vectors as a Mixed-Integer Non-Convex Programming (MINCP) problem, where the optimal solution is probably intractable. Consequently, important motivations for developing the algorithms are to provide a range of near-optimal results and proper benchmarking for the ISAC-MSC system.

C. Contributions

The main contributions of the work are summarized as follows.

1) We propose an ISAC-MSC secure communication system and formulate the optimization problem maximizing the target sensing SNR under the constraints of transmit power budget and the number for collaborative LEO satellites. By alternately optimizing two variables, i.e., BF vectors and LEO satellite assignment, we decompose the optimization problem into two sub-problems. We also discuss the non-convexity of the optimization problem.
2) We design improved Continuous Particle Swarm (CP) optimization and Discrete Particle Swarm (DP) optimization algorithms incorporating security exclusion terms to address the optimization problems in LEO satellite assignment. We also develop the Power Approximation (PA) optimization, Inner Approximation (IA) optimization, and Joint Sensing and Communication BF (JSC-BF) optimization algorithms. We give the complexity analysis of algorithms.
3) The traditional Shortest Path (SHP) algorithm served as the comparative baseline, and the proposed DP and CP based algorithm significantly outperforms it in target sensing SNR. Compared to the single satellite system, the proposed ISAC-MSC system has approximate linear enhancement with increasing transmit power budget in terms of the sensing performance. We evaluate the effect of LEO satellites' altitude on the target sensing SNR, proving that the ISAC-MSC system is highly reliable.
4) From the viewpoint of secure communication, we study the communication secrecy rate of the various algorithms proposed. The JSC-BF algorithms still provide excellent system security. We verify that the number of transmitting antennas has a considerable impact on the security improvement. We confirm the communication

TABLE I
NOTIONS

Notation	Description
$\mathcal{M} = \{m\}$	LEO satellite set
$\mathcal{K} = \{k\}$	Ground user set
$\mathcal{J} = \{j\}$	Eavesdropper set
i	Index of time slots
δ_i	The length of the time slot i
T	Number the time slots
$\mathcal{M}_t$	Transmitter LEO satellite set
$\mathcal{M}_r$	Receiver LEO satellite set
$\mathcal{M}_t^k$	Transmitter LEO satellite and user matching set
$\mathcal{M}_r^j$	Receiver LEO satellite and eavesdropper matching set
$R_k[i]$	The communication rate of user k at time slot i
$R_j[i]$	The transmission rate of eavesdropper j at time slot i
$\gamma_s[i]$	The target sensing SNR at time slot i
$\mathbf{h}_k$	Channel vector of user k
$\mathbf{w}_k$	BF vector of user k form all LEO satellites
P_m	Power budget per LEO satellite
M_0	Number of satellites served per user
N_t	Number of BF transmitting antennas
N_r	Number of BF receiving antennas

secrecy rate appearing inflection point and stabilization with an increasing power budget in the case of imperfect MSC synchronization.

D. Organization and Notations

The rest of this paper is organized as follows. Section II introduces an ISAC-MSC security communication system model. Section III proposes an optimization problem under the constraints of power and the number of collaborative LEO satellites. In Sections IV and V, we designed the algorithms by jointly optimizing BF vectors and LEO satellite assignments. Section VI presents numerical results to verify the superiority of the proposed algorithms, and the work is concluded in Section VII.

The key symbols in this paper are defined as follows: The operator $|\cdot|$ represents the absolute value of a complex number, while $||\cdot||$ denotes the norm. $(\cdot)^H$ refers to the conjugate transpose of the matrix. $(\cdot)^T$ and $(\cdot)^\dagger$ indicate the transpose and pseudo-inverse operation. $\det(\cdot)$ represents the determinant of a matrix. $\mathbb{C}$ and $\mathbb{R}$ denote the set of complex and real numbers, respectively. $[x]^+$ and x^* denote $\max\{0, x\}$ and conjugate operators. $\mathbb{E}(\cdot)$ is the statistical expectation. $\|\cdot\|^2$ represents the square of the Euclidean norm.

The frequently utilized notations in this work are first listed in Table I.

II. SYSTEM MODEL

In this paper, we consider an ISAC-MSC secure communication system, as shown in Fig. 1, which consists of M LEO satellites, K communications users, and J eavesdroppers.

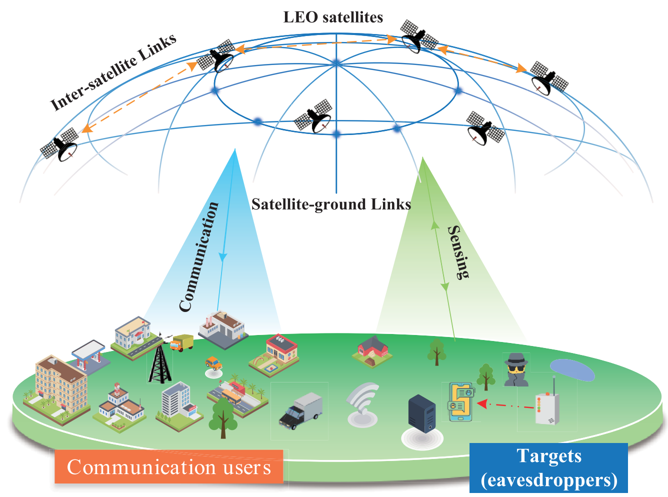

Fig. 1. An ISAC-MSC secure communication system model, where LEO satellites collaboratively transmit ISAC signals to users and targets, sensing eavesdroppers, and establishing secure communications. Multiple eavesdroppers attempt to capture the users' information.

More specifically, M LEO satellites are divided into transmitter set $\mathcal{M}_t = \{1, \ldots, m_t, \ldots M_t\}$ and receiver set $\mathcal{M}_r = \{1, \ldots, m_r, \ldots M_r\}$, where the satellites are distributed in orbits at different altitudes.

In the downlink, the matching set $\mathcal{M}_t^k$ represents an LEO satellite m_t transmitting ISAC signals to a user k. At the same time, another matching set $\mathcal{M}_r^j$ is an LEO satellite m_r receiving reflections/scattering signals from an eavesdropper j, which is the sensed target.[1] As a further development, the satellite scheduling period W_t is divided into T equal time slots. The length of the time slot i is defined as $\delta_i = W_t/T$. The set of time slots is denoted $\mathcal{W}$. Note that the antennas of LEO satellites are classified as N_t transmitting antennas or N_r receiving antennas based on the role of the satellite.[2] Both users and eavesdroppers are equipped with a single antenna. We consider that LEO satellites can be jointly designed and planned to transmit/receive ISAC signals simultaneously.

Next, the transmitter LEO satellites transmit K communication signals and J sensing signals at the time slot i, defined as $x_k[i], k \in \mathcal{K}$ and $x_j[i], j \in \mathcal{J}$, respectively. Let $\mathcal{K} \triangleq \{1, \ldots, K\}$ and $\mathcal{J} \triangleq \{1, \ldots, J\}$ indicate the communication and sensing signals, respectively. The set of all signals is denoted as $\mathcal{Q} = \{1, \ldots, K, K+1, \ldots, Q\}$, where $Q = K + J$. Accordingly, the signal transmitted by the LEO

[1] Given that eavesdroppers in high-risk areas usually employ high-gain mobile devices to obtain information, we have included both large fixed antennas and vehicle-mounted eavesdropping devices within the scope of potential eavesdropping equipment [26], [27], [28]. While geographically distinct, the users and eavesdroppers are located in relative proximity to each other.

[2] LEO satellites perform communication and sensing at the same moment, where LEO satellites with different functions accomplish these two tasks. Currently, constellations that possess MSC capabilities and have formulated ISAC implementation plans include Starlink, OneWeb, and Tiansuan Constellation, among others.

satellite m_t at the time slot i is given as

$$\mathbf{x}_{m_t}[i] = \sum_{k\in\mathcal{K}} \mathbf{w}_{m_t,k} x_k[i] + \sum_{j\in\mathcal{J}} \mathbf{w}_{m_t,j} x_j[i] = \sum_{q\in\mathcal{Q}} \mathbf{w}_{m_t,q} x_q[i], \tag{1}$$

where $x_q[i]$ is the q-th signal at the time slot i, $\mathbf{w}_{m_t,q} \in \mathbb{C}^{N_t\times 1}$ represents the BF vectors for the q-th signal applied by the LEO satellite m_t, and $x_q[i]$ is normalized unity power signals, i.e., $\mathbb{E}\left[|x_q[i]|^2\right] = 1$. The BF vectors are constrained by the satellite power P_m, with $\mathbb{E}\left[\|\mathbf{x}_{m_t}[i]\|^2\right] = \sum_{q\in\mathcal{Q}} \|\mathbf{w}_{m_t,q}\|^2 \leq P_m$. The BF vectors of all satellite signals for user k are stacked as $\mathbf{w}_q = \left[\mathbf{w}_{q,1}^T, \ldots, \mathbf{w}_{M_t,q}^T\right]^T$. We consider that the information from the sensing and communication signals is statistically independent [6].

Furthermore, only the line-of-sight component is considered between the user and the LEO satellite. For the satellite-terrestrial communication link, a well-known free-space path loss is introduced in this paper. Therefore, the channel model between the LEO satellite m_t and user k is expressed as [23], [29], [30]

$$\mathbf{h}_{m_t,k} = \sqrt{G_t G_r^k} \sqrt{\left[\frac{c}{(f_c + \bar{v})(4\pi d_{m_t,k})}\right]^2} \mathbf{a}(\phi_{m_t}), \tag{2}$$

where $\bar{v} = v^{\text{sat}} + v^{\text{gro}}$ represents the Doppler frequency shift of the direct line-of-sight link coming from the mobility of the LEO satellites and ground terminals, where c and f_c stands for the speed of light and the central frequency. $d_{m_t,k}$ represents distance from the satellite m_t to the user k, respectively. Furthermore, G_t and G_r^k are the satellite antenna transmit gain and user k receive antenna gain, respectively. $\mathbf{a}(\phi_{m_{t,k}}) = \frac{1}{\sqrt{N_t}}\left[1 e^{-j\frac{2\pi s(f_c+\bar{v})}{c}\sin\phi_{m_{t,k}}} \ldots e^{-j(N_t-1)\frac{2\pi s(f_c+\bar{v})}{c}\sin\phi_{m_{t,k}}}\right]$ is the array response vector, where ϕ and s are the relative angle of arrival between nodes and antenna spacing. $\mathbf{h}_k = \left[\mathbf{h}_{1,k}^T, \ldots, \mathbf{h}_{M_t,k}^T\right]^T$ is defined as an equivalent compressed channel.

Thus, the received signal model of user k at the time slot i is computed as

$$y_k[i] = \mathbf{I}_{\text{DS}} + \mathbf{I}_{\text{MUI}} + \mathbf{I}_{\text{SI}} + n_k[i] = \sum_{m_t\in\mathcal{M}_t} \mathbf{h}_k^H \mathbf{x}_{m_t}[i] + n_k[i], \tag{3}$$

where

$$\mathbf{I}_{\text{DS}} = \mathbf{h}_k^H \mathbf{w}_k x_k[i], \tag{4}$$

$$\mathbf{I}_{\text{MUI}} = \sum_{k'\in\mathcal{K}\setminus\{k\}} \mathbf{h}_{k,m_t}^H \mathbf{w}_{k',m_t} x_{k'}[i], \tag{5}$$

and

$$\mathbf{I}_{\text{SI}} = \sum_{j\in\mathcal{J}} \mathbf{h}_k^H \mathbf{w}_j x_j[i], \tag{6}$$

where $\mathbf{I}_{\text{DS}}$, $\mathbf{I}_{\text{MUI}}$ and $\mathbf{I}_{\text{SI}}$ represent desired signal, multi-user interference and sensing signal interference, respectively. $n_k[i] \sim \mathcal{CN}(0, \sigma_k^2)$ is the Gaussian white noise of user k, and σ_k^2 represents noise power of user k. Then, the communication rate for user k can be written as

$$R_k[i] = \log_2\left(1 + \frac{\left|\mathbf{h}_k^H \mathbf{w}_k\right|^2}{\sum_{k'\in\mathcal{K}\setminus\{k\}} \left|\mathbf{h}_k^H \mathbf{w}_{k'}\right|^2 + \sum_{j\in\mathcal{J}} \left|\mathbf{h}_k^H \mathbf{w}_j\right|^2 + \sigma_k^2}\right). \tag{7}$$

Similarly, the received signal of the eavesdropper j can be expressed as

$$y_j[i] = \sum_{k\in\mathcal{K}} \mathbf{h}_j^H \mathbf{w}_k x_k[i] + n_j[i], \tag{8}$$

where $\mathbf{h}_j$ is defines the stacked channel vector for the eavesdropper j, and $n_j[i] \sim \mathcal{CN}(0, \sigma_j^2)$ is the receiver noise of eavesdropper, and σ_j^2 denotes noise power about eavesdropper j.

To evaluate communication and sensing performance, assuming the worst-case scenario in which an eavesdropper can receive signals of interest, the rate of eavesdropper j for the user k is given as

$$R_j[i] = \log_2\left(1 + \frac{\left|\mathbf{h}_j^H \mathbf{w}_k\right|^2}{\sigma_j^2}\right). \tag{9}$$

By transmitting the dual-functional waveforms to sense the eavesdropper j, the signal received by the LEO satellite m_r at the time slot i is obtained as

$$\tilde{\mathbf{y}}_{m_r}[i] = \sum_{m_t\in\mathcal{M}_t} \kappa\sqrt{\mu}\,\mathbf{a}(\phi_{m_r})\,\mathbf{a}^H(\phi_{m_t})\,\mathbf{x}_{m_t}[i] + \mathbf{n}_{m_r}[i], \tag{10}$$

where $\kappa \sim \mathcal{CN}(0, \sigma_{m_r,m_t}^2)$ stands for the RCS of eavesdropper j;

$$\mu = \frac{\lambda^2}{(4\pi)^3 R_{m_r,j}^2 R_{m_t,k}^2}, \tag{11}$$

is the overall gain of the downlink sensing channel and backhaul channel; $R_{m_t,k}$ and $R_{m_r,j}$ represent the distances from user k served satellite m_t to the eavesdropper j and eavesdropper j to the satellite m_r, respectively. λ is denoted as the wavelength. $\mathbf{n}_{m_r}[i] \sim \mathcal{CN}(0, \sigma_j^2 \mathbf{I}_{N_r})$ is the receiver noise at the satellite m_r for eavesdropper j [23].

As a result, the joint sensing SNR can be derived as

$$\gamma_s[i] = \frac{\sum_{m_t\in\mathcal{M}_t}\sum_{m_r\in\mathcal{M}_r} \mu\sigma_{m_r,m_t}^2 \left\|\mathbf{a}^H(\phi_{m_t})\,\mathbf{W}_{m_t,q}\right\|^2}{M_0\sigma_{\text{rad}}^2}, \tag{12}$$

where M_0 represents the number of LEO satellite collaborations; $\mathbf{W}_{m_t,q} \triangleq [\mathbf{w}_{m_t,1}, \ldots, \mathbf{w}_{m_t,q}, \ldots, \mathbf{w}_{m_t,Q}]$ is defined as concatenating the BF vectors of all the communication users and the sensing target. σ_{rad}^2 is radar receiver noise power.

III. Problem Formulation and Analysis

A. Problem Formulation

We formulate the optimization problem to maximize target sensing SNR by jointly optimizing sensing BF vectors, communication BF vectors, and LEO satellites assignment.

The objective function incorporates the average sensing level among T time slots, which is formulated as

$$\max_{\Omega_i,\mathbf{w}} \sum_{i\in\mathcal{W}} \gamma_s[i] \tag{13a}$$

$$\text{s.t.} \min_{j\in\mathcal{J}} \left(\sum_{i\in\mathcal{W}} R_k[i] - R_j[i]\right) \geq \Gamma_{\text{SR}}, \forall k \in \mathcal{K}, \tag{13b}$$

$$\|\mathbf{w}_{m_t}\|^2 \leq P_m, \forall m_t \in \mathcal{M}_t, \tag{13c}$$

$$\Omega_i = \left\{\mathcal{M}_t^k, \mathcal{M}_r^j\right\}, \forall k \in \mathcal{K}, \forall j \in \mathcal{J}, \tag{13d}$$

$$\left|\mathcal{M}_t^k\right| = \left|\mathcal{M}_r^j\right| = M_0, \forall k \in \mathcal{K}, \forall j \in \mathcal{J}, \tag{13e}$$

where the constraint (13b) represents the secrecy rate for any user k that is greater than the threshold Γ_{SR}, and (13c) is the power constrain at each transmitting LEO satellite, reflecting that the transmit power from the LEO satellites m_t does not exceed the power threshold P_m. In (13d), we introduce Ω_i as a set that includes all the matching LEO satellites, e.g., transmitter satellites and receiver satellites. The constraint (13e) confines the number of transmitter and receiver satellites to be equal to M_0.

The optimization decisions regarding the BF vectors and LEO satellites assignment are coupled with each other in (13). It is generally accepted that jointly optimizing the two variables is challenging. Designing the BF vectors within each time slot depends on the strategy of LEO satellite assignment, while achieving an optimal LEO satellite assignment needs to rely on the optimization results for BF vectors. Therefore, in order to obtain an overall solution for the ISAC-MSC system, it is necessary to capture the dependencies of the variables and design the optimization algorithm with alternating iterations. On the other hand, the optimization problem (13) is an MINCP problem due to the nonlinear and non-convex functions in (13d), (13e), and (13b). Next, we design alternating iterative algorithms to optimize the LEO satellites assignment by fixing the BF vectors, obtaining the set Ω_i, and then optimizing $\mathbf{w}$, and finally converging to an optimal solution.

B. Physical Relative Motion in MSC

Due to the effect of the earth's curvature on large-scale LEO satellites, we establish the visible constraints between the LEO satellite and the user or target. As shown in Fig. 2, for a satellite and a user to be visible to each other, the user should be within the coverage area of the satellite. When the user is located at the edge of the coverage area, the elevation angle is minimum, defined as β_k. According to the law of sines, the functional relationship formed by the terminal, satellite and center is

$$\frac{R_{\text{earth}} + l_m}{\sin(90^\circ + \beta_k)} = \frac{R_{\text{earth}}}{\sin(\zeta)}. \tag{14}$$

Using the triangle angle sum theorem, by rearranging we obtain

$$\cos(\beta_k + \partial_{\max}) = \frac{R_{\text{earth}} \cdot \cos(\beta_k)}{R_{\text{earth}} + l_m}, \tag{15}$$

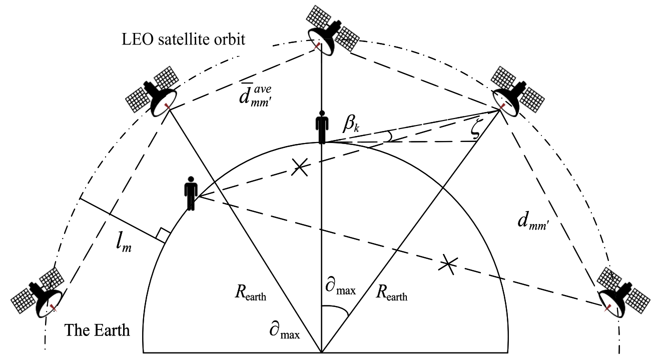

Fig. 2. Visible relationship between LEO satellites and users. LEO satellites are distributed at different orbital altitudes, where the orbit's vertical distance from the ground is given as l_m. ζ is the auxiliary angle. With the rapid movement of the LEO satellites, the distances among them cannot be less than $\bar{d}_{mm'}^{ave}$.

After taking the arccos and collating the results, the geocentric angle between the satellite and user reaches the maximum of $\partial_{\max}$ as [31]

$$\partial_{\max} = \arccos\left(\cos\beta_k \times \frac{R_{\text{earth}}}{R_{\text{earth}} + l_m}\right) - \beta_k, \tag{16}$$

where l_m represents the orbital altitude of LEO satellites.

If the geocentric angle between the LEO satellite and the user is less than or equal to the maximum value $\partial_{\max}$, they are visible to each other. Define $\vartheta_{m_t k}$ be the geocentric angle between LEO satellite m_t and user k. Let a binary variable $\xi_{m_t k}$ represent whether the user is visible to the LEO satellites or not, being given as

$$\xi_{m_t k} = \begin{cases} 1, & \text{if } \vartheta_{\mathrm{m_t k}} \leq \partial_{\max}; \\ 0, & \text{otherwies} \end{cases} \tag{17}$$

To ensure that the ISAC-MSC can improve the security while achieving the goal of large coverage, the distance between the inter-satellite links in orbit is taken into account, where the distance between two satellites is derived as

$$d_{mm'} = \sqrt{(\hat{x}_m - \hat{x}_{m'})^2 + (\hat{y}_m - \hat{y}_{m'})^2 + (\hat{z}_m - \hat{z}_{m'})^2}, \tag{18}$$

where $(\hat{x}_m, \hat{y}_m, \hat{z}_m)$ and $(\hat{x}_{m'}, \hat{y}_{m'}, \hat{z}_{m'})$ stand for the three-dimensional coordinates of the two LEO satellites.

In the ISAC-MSC model, the average distance $\bar{d}_{mm'}^{ave}$ is calculated as

$$\bar{d}_{mm'}^{ave} = \frac{\int_0^{\delta_i} d_{mm'}[i]\, di}{\delta_i}, \tag{19}$$

which ensures the effectiveness of MSC by limiting the average distances between LEO satellites. Based on the above derivation, the visible matrix between the LEO satellites and users is obtained as

$$\psi_{M_t \times K} = \begin{bmatrix} \xi_{11} & \xi_{12} & \cdots & \xi_{1K} \\ \xi_{21} & \xi_{22} & \cdots & \xi_{2K} \\ \cdots & \cdots & \xi_{m_t k} & \cdots \\ \xi_{M_t 1} & \xi_{M_t 2} & \cdots & \xi_{M_t K} \end{bmatrix}_{M_t \times K}. \tag{20}$$

IV. LEO Satellite-User Assignment Optimization

In this section, we design an improved Particle Swarm Optimization (PSO) algorithm for LEO satellite assignment by fixing BF vectors.

A. Improved Discrete PSO Algorithm

An optimal LEO satellite assignment should ensure that the elements of the system are visible to each other. Thus, ξ_{m_tk} in (20) can be regarded as a condition for the variable Ω_i of (13a). The optimization problem regarding user k in Ω_i is transformed can be further for optimization ξ_{m_tk}.[3]

For dynamic resource matching in LEO satellites, we propose an enhanced improved discrete PSO algorithm, incorporating an exclusion term to guide the particle swarm proactively away from regions that fail to satisfy security constraints, thereby enabling more efficient exploration of optimal solutions within the feasible domain. In the ISAC-MSC system, the discrete LEO satellite assignment problem is mapped to particle motion space.

The DP algorithm is based on the concepts of population and evolution, and it achieves the search for optimal solutions in complex space through collaboration and competition among individuals. Similarly, for the ISAC-MSC system, obtaining higher sensing SNR requires more self-knowledge and group-knowledge to form an optimal LEO satellite-user/target assignment strategy. Particles have only two attributes: velocity and position, where velocity represents the speed of movement and position represents the direction of movement [32]. The particles take on only two values 0 and 1, in the state space. Meanwhile, each value of the particle's velocity represents the possibility of the particle's position taking the value of 0 or 1. For finding the optimal values, the particles update their own velocity according to

$$\begin{aligned} v_{su}(e+1) &= \varpi v_{su}(e) + c_1 r_1(e)\left[\rho_{su}(e) - \chi_{su}(e)\right] \\ &\quad + c_2 r_2(e)\left[\rho_u^g(e) - \chi_{su}(e)\right] \\ &\quad - c_3 r_3(e)\left[\rho_u^w(e) - \chi_{su}(e)\right], \end{aligned} \tag{21}$$

where ϖ represents the inertial weights, c_1 and c_2 are the individual cognitive and social cognitive learning factors, respectively. e is defined as the number of iterations, and r_1 and r_2 are uniform random numbers in the range of $[0, 1]$, which increase the randomness of particles' flights. ρ_{su} and ρ_u^{g} are the optimal solution found by the individual and optimal solution found by the global, respectively. χ_{su} stands for the current position of particles [33]. At each iteration, besides seeking the globally optimal solution ρ_u^g that satisfies the constraints, a 'globally worst-case security solution' $\rho_u^w(e)$ is concurrently recorded. The physical significance of $\rho_u^w(e)$ is the position of the particle with the highest perceived SNR among all particles currently failing to meet the secure rate threshold. c_3 represents the security exclusion factor, governing the determination and velocity of particles escaping from unsafe constraints. $r_3(e)$ denotes a random number within $[0, 1]$, introducing randomness to the exclusion behavior.

[3]The matrix $\mathcal{M}_r^j$ is updated to consist of ξ_{m_tk}. The matching of target j and receiver satellite m_r is solved in the same way.

To realize the detection of particles faster in the early stages and more accurately, the dynamic linear decreasing weights strategy is used as

$$\varpi = \varpi_{\max} - \frac{(\varpi_{\max} - \varpi_{\min}) \cdot e}{E_{\max}}, \tag{22}$$

where $E_{\max}$ denotes the maximum number of iterations. $\varpi_{\max}$ and $\varpi_{\min}$ are the maximum and minimum inertia weights, respectively.

At the same time, the position of the particles is updated as

$$f(v_{su}) = \frac{1}{1 + \exp(-v_{su})}. \tag{23}$$

Furthermore, the particle position result is given as

$$\chi_{su} = \begin{cases} 1, & \text{if } r_3 < f(v_{su}); \\ 0, & \text{otherwise} \end{cases}, \tag{24}$$

where r_3 is a random number from 0 to 1.

We pre-process the DP algorithm to search the target space and incorporate a penalty function to ensure secure communication in the MSC systems. Thus, the optimization problem is reformulated as

$$\max_{\Omega_i} \sum_{i \in \mathcal{W}} \gamma_s[i] + \varphi\left[\Gamma_{\mathrm{SR}}^{\gamma} - \min_{j \in \mathcal{J}}(\gamma_k[i] - \gamma_j[i])\right]^+ \tag{25a}$$

$$\text{s.t.} \quad \Omega_i = \left\{\mathcal{M}_t^k, \mathcal{M}_r^j\right\}, \forall k \in \mathcal{K}, \forall j \in \mathcal{J}, \tag{25b}$$

$$\left|\mathcal{M}_t^k\right| = \left|\mathcal{M}_r^j\right| = M_0, \forall k \in \mathcal{K}, \forall j \in \mathcal{J}, \tag{25c}$$

$$\varphi < 0, \tag{25d}$$

where φ is the penalty factor. The first step of the DP algorithm process is initializing the particle swarm-related parameters and penalty factors while calculating the target value for each particle. Then, for each particle's individual extreme value and global extreme are compared with the target value to update the particle's velocity and position. Finally, calculating the result of (13a) in each iteration is updated until convergence. The complexity of DP algorithm is $\mathcal{O}(N_p E_{\max} MKJ \times T \max(\gamma_s))$, where N_p represents the number of particles [34].

B. An Efficient Continuous PSO Algorithm

While the DP algorithm is suitable for the discrete nature of satellite assignment, we can often achieve faster convergence and a more refined search by relaxing the problem into a continuous domain. In this subsection, we propose a more efficient CP algorithm to solve for the satellite assignment. More specifically, the discrete variable ξ_{m_tk} of (20) is made continuous. Define matrix $\wp$ be the LEO satellite assignment matrix with respect to the continuous variable $\bar{\xi}_{m_tk} \in [0, 1]$. The value closer to 1 indicates a stronger preference for that assignment. The collection of these weights forms the assignment matrix

$$\begin{aligned} \psi[i+1] &= \psi[i] \cdot \wp[i] \\ &= \begin{bmatrix} \bar{\xi}_{11} & \bar{\xi}_{12} & \cdots & \bar{\xi}_{1K} \\ \bar{\xi}_{21} & \bar{\xi}_{22} & \cdots & \bar{\xi}_{2K} \\ \cdots & \cdots & \bar{\xi}_{m_tk} & \cdots \\ \bar{\xi}_{M_t1} & \bar{\xi}_{M_t2} & \cdots & \bar{\xi}_{M_tK} \end{bmatrix}_{M_t \times K}. \end{aligned} \tag{26}$$

In the CP algorithm, the position of each particle, χ_{su} corresponds to the flattened vector of the assignment matrix ψ. The particles explore the solution space by updating their velocity and position in each iteration. Compared to the DP algorithm, the CP algorithm often employs a compression factor to control the particle's velocity, which effectively balances the local and global search capabilities and ensures convergence. The velocity is updated as

$$\begin{aligned} v_{su}(e+1)=&\varpi v_{su}(e)+\ell\left\{c_1r_1(e)\left[\rho_{su}(e)-\chi_{su}(e)\right]\right\}\\ &+c_2r_2(e)\left[\rho_u^{\mathrm{g}}(e)-\chi_{su}(e)\right]\\ &-c_3r_3(e)\left[\rho_u^{w}(e)-\chi_{su}(e)\right], \end{aligned} \tag{27}$$

where the compression factor ℓ is defined as

$$\ell=\frac{2}{\left|2+4c-\sqrt{2c^2-4c}\right|}, \tag{28}$$

with $c=c_1+c_2$. The corresponding particle update position is

$$\chi_{su}(e+1)=\chi_{su}(e)+v_{su}(e+1). \tag{29}$$

After the iterative process converges, the resulting continuous weight matrix ψ is converted back to a discrete assignment decision. For each user k, we select the M_0 satellites with the highest assignment weights $\bar{\xi}_{m_tk}$ to form the final assignment set $\mathcal{M}_t^k$. The complexity of CD algorithm is $\mathcal{O}\left(\left(M_t^N+M_r^N\right)N_pE_{\max}KJ\times T\max\left(\gamma_s\right)\right)$, M_t^N are M_r^N the particle dimensions with respect to transmitter and receiver satellites, respectively [34]. The main source of algorithmic complexity is the number of particle swarms and iterations for solving the optimization problem (13).

V. Optimization Approaches for BF Vectors

Considering the difficulties in solving non-convex problems with BF vectors, the PA, IA and JSC-BF algorithms are designed based on the utilization of the above-obtained LEO satellite assignment results. Hence, the optimization problem with respect to $\mathbf{w}$ is reformulated as

$$\max_{\mathbf{w}} \sum_{i\in\mathcal{W}}\gamma_s[i] \tag{30a}$$

$$\text{s.t. } \min_{j\in\mathcal{J}}\left(\sum_{i\in\mathcal{W}}R_k[i]-R_j[i]\right)\geq\Gamma_{\mathrm{SR}},\forall k\in\mathcal{K}, \tag{30b}$$

$$\left\|\mathbf{w}_{m_t}\right\|^2\leq P_m,\forall m_t\in\mathcal{M}_t. \tag{30c}$$

A. BF Vectors Optimization Algorithm via Power Approximation

In the subsection, we design the PA algorithm, where the power allocation for sensing and communication is first optimized. The communication or sensing power of the entire antenna array for a LEO satellite can be expressed by summing the power from all units. Therefore, converting the BF vectors to power p, the target sensing SNR with p is obtained as

$$\gamma_s[i]=\frac{\sum\limits_{m_t\in\mathcal{M}_t}\sum\limits_{m_r\in\mathcal{M}_r}\mu\sigma_{m_r,m_t}^2\left(p_s+p_c\right)}{M_0\sigma_{\mathrm{rad}}^2}, \tag{31}$$

where p_s is the power allocated for the sensing beam. p_c is the power allocated for the communication beam.

To handle the non-convexity of (30b), the user's rate is approximated as (32), shown at the bottom of the page, where $p_c(\varepsilon)$ is the value of p_c in the ε-th iteration, and

$$\begin{aligned} Z(p)=&\left\|\mathbf{h}_k\right\|^2\Bigg(\left\|\mathbf{h}_k\right\|^2p_c^k(\varepsilon)+\sigma_k^2+\sum_{k'\in\mathcal{K}\setminus\{k\}}\left\|\mathbf{h}_k\right\|^2p_c^{k'}\\ &+\sum_{j\in\mathcal{J}}\left\|\mathbf{h}_k\right\|^2p_c^j\Bigg)^{-1}(\ln 2)^{-1}. \end{aligned} \tag{33}$$

In addition, the rate of eavesdropper j is recalculated as

$$R_j(p)=\log_2\left(1+\frac{\left\|\mathbf{h}_k\right\|^2p_c^k}{\sigma_j^2}\right). \tag{34}$$

The set of transmit power from the satellite is defined as $\mathbf{p}=\{p_c,p_s\}$. Based on the above description, the optimization problem on power is formulated as

$$\max_{\mathbf{p}} \sum_{i\in\mathcal{W}}\gamma_s(p) \tag{35a}$$

$$\text{s.t. } \min_{j\in\mathcal{J}}\left(\sum_{i\in\mathcal{W}}\bar{R}_k(p)-R_j(p)\right)\geq\Gamma_{\mathrm{SR}},\forall k\in\mathcal{K}, \tag{35b}$$

$$p_c+p_s\leq P_m, \tag{35c}$$

$$p_c\geq 0, p_s\geq 0, \tag{35d}$$

which can be efficiently solved using the CVX solver [35]. Meanwhile, the optimal value is gradually approximated based on the results of each iteration.

The sensing BF vectors can be constructed as

$$\mathbf{w}_{m_t}^s=\sqrt{p_s}\frac{\left(\mathbf{I}-\mathbf{H}_{m_t}\left(\mathbf{H}_{m_t}^H\mathbf{H}_{m_t}\right)^{\dagger}\mathbf{H}_{m_t}^H\right)\mathbf{a}\left(\theta_{m_t}\right)}{\left\|\left(\mathbf{I}-\mathbf{H}_{m_t}\left(\mathbf{H}_{m_t}^H\mathbf{H}_{m_t}\right)^{\dagger}\mathbf{H}_{m_t}^H\right)\mathbf{a}\left(\theta_{m_t}\right)\right\|}, \tag{36}$$

$$\begin{aligned} R_k(p)=&\log_2\left(1+\frac{\left\|\mathbf{h}_k\right\|^2p_c^k}{\sum\limits_{k'\in\mathcal{K}\setminus\{k\}}\left\|\mathbf{h}_k\right\|^2p_c^{k'}+\sum\limits_{j\in\mathcal{J}}\left\|\mathbf{h}_k\right\|^2p_c^j+\sigma_k^2}\right)\leq\bar{R}_k(p)\\ =&\log_2\left(1+\frac{\left\|\mathbf{h}_k\right\|^2p_c^k(\varepsilon)}{\sum\limits_{k'\in\mathcal{K}\setminus\{k\}}\left\|\mathbf{h}_k\right\|^2p_c^{k'}+\sum\limits_{j\in\mathcal{J}}\left\|\mathbf{h}_k\right\|^2p_c^j+\sigma_k^2}\right)+Z(p)\left(p_c^k-p_c^k(\varepsilon)\right). \end{aligned} \tag{32}$$

where $\mathbf{I}$ is the unit matrix, and $\mathbf{H}_{m_t}$ is defined as concatenating the channels from the transmitter satellite m_t to all users.

The communication BF vectors can be written as [36]

$$\mathbf{w}_{m_t}^{c}=\sqrt{\frac{p_c}{N_t}}\mathbf{a}\left(\phi_{m_t,q}\right). \tag{37}$$

The complexity of PA algorithm is mainly due to the number of ε iterations.

B. BF Vectors Optimization Algorithm by Inner Approximation

In this subsection, we address the non-convexity of (30) by relaxing the target sensing SNR and secrecy rate constraints and construct the IA algorithm. In what follows, we obtain an inner approximation of $\left\|\mathbf{a}^{H}\left(\phi_{m_t}\right)\mathbf{W}_{m_t,q}\right\|^{2}$ at the ι-th iteration by [37] (38), shown at the bottom of the page. Next, define an inner approximation of $\gamma_s^{\iota}(\mathbf{w})$ at the ι-th iteration. [SETCOUNTER-OBJ:

To deal with the non-concave constraints in (30b), the rate of user k is approximated as

$$R_k^{\iota}(\mathbf{w})\triangleq A_k^{\iota}+2\mathbb{R}\left\{\mathbf{B}_k^{\iota}\mathbf{w}_k\right\}-\sum_{\bar{j}\in\{\mathcal{K},\mathcal{J}\}\backslash\{k\}}\left(\mathbf{w}_{\bar{j}}\right)^{H}\mathbf{C}_k^{\iota}\mathbf{w}_{\bar{j}}, \tag{39}$$

where

$$A_k^{\iota}=R_k(\mathbf{w}^{\iota})-\frac{\left|\mathbf{h}_k^{H}\mathbf{w}_k^{\iota}\right|^{2}+\sigma_k^{2}}{\sum\limits_{\bar{j}\in\{\mathcal{K},\mathcal{J}\}\backslash\{k\}}\left|\mathbf{h}_k^{H}\mathbf{w}_{\bar{j}}^{\iota}\right|^{2}+\sigma_k^{2}}+\frac{\sigma_k^{2}}{\sum\limits_{\bar{j}\in\{\mathcal{K},\mathcal{J}\}}\left|\mathbf{h}_k^{H}\mathbf{w}_{\bar{j}}^{\iota}\right|^{2}+\sigma_k^{2}}, \tag{40}$$

$$\mathbf{B}_k^{\iota}=\frac{\left(\mathbf{h}_k^{H}\mathbf{w}_k^{\iota}\right)^{*}\mathbf{h}_k^{H}}{\sum\limits_{\bar{j}\in\{\mathcal{K},\mathcal{J}\}\backslash\{k\}}\left|\mathbf{h}_k^{H}\mathbf{w}_{\bar{j}}^{\iota}\right|^{2}+\sigma_k^{2}}, \tag{41}$$

and

$$\mathbf{C}_k^{\iota}=\left|\mathbf{h}_k^{H}\mathbf{w}_k^{\iota}\right|^{2}\mathbf{h}_k\mathbf{h}_k^{H}\left(\sum_{\bar{j}\in\{\mathcal{K},\mathcal{J}\}}\left|\mathbf{h}_k^{H}\mathbf{w}_{\bar{j}}^{\iota}\right|^{2}+\sigma_k^{2}\right)^{-1}\times\left(\sum_{\bar{j}\in\{\mathcal{K},\mathcal{J}\}\backslash\{k\}}\left|\mathbf{h}_k^{H}\mathbf{w}_{\bar{j}}^{\iota}\right|^{2}+\sigma_k^{2}\right)^{-1}. \tag{42}$$

Proof: Appendix A. ■

With (38) and (39), the optimization problem (30) is transformed into a convex quadratic problem as

$$\max_{\mathbf{w}}\sum_{i\in\mathcal{W}}\gamma_s^{\iota}(\mathbf{w}) \tag{43a}$$

$$\text{s.t.}\min_{j\in\mathcal{J}}\left(\sum_{i\in\mathcal{W}}R_k^{\iota}(\mathbf{w})-R_j(\mathbf{w})\right)\geq\Gamma_{\mathrm{SR}},\forall k\in\mathcal{K}, \tag{43b}$$

$$\sum_{k\in\mathcal{K}}\left\|\mathbf{w}_{m_t,k}\right\|^{2}\leq P_m,\forall m_t\in\mathcal{M}_t, \tag{43c}$$

which is a tight maximization function with $\mathbf{w}^{\iota}$.

Furthermore, the sensing BF vectors are initialized first through (36), which eliminates the interference of the sensing beam to the communication beam. Then, by fixing the sensing BF vectors, the optimization problem of max-min secrecy rate is formulated to solve with the threshold Γ_{SR} as

$$\max_{\mathbf{w}}\min_{j\in\mathcal{J}}\left(\sum_{i\in\mathcal{W}}R_k^{\iota}(\mathbf{w})-R_j(\mathbf{w})\right),\forall k\in\mathcal{K} \tag{44a}$$

$$\text{s.t.}\sum_{k\in\mathcal{K}}\left\|\mathbf{w}_{m_t,k}\right\|^{2}\leq p_c,\forall m_t\in\mathcal{M}_t. \tag{44b}$$

Since (44a) is a non-convex problem, we need to approximate the optimum with the results of each iteration. The non-convex problem of (44a) is solved by utilizing a tight max-min function at the ι-th iteration as

$$\max_{\mathbf{w}}\min_{j\in\mathcal{J}}\left(\sum_{i\in\mathcal{W}}R_k(\mathbf{w})-R_j(\mathbf{w})\right),\forall k\in\mathcal{K} \tag{45a}$$

$$\text{s.t.}\sum_{k\in\mathcal{K}}\left\|\mathbf{w}_{m_t,k}\right\|^{2}\leq p_c,\forall m_t\in\mathcal{M}_t. \tag{45b}$$

Algorithm 1 IA

Require: $\mathbf{w}_{m_t}^{c}$, $\mathbf{w}_{m_t}^{s}$, p_s, p_c, $\iota=0$, T, K.
Ensure: $\mathbf{w}_q$

repeat
for $i=1,\ldots,T$ **do**
for $k=1,\ldots,K$ **do**
Solve (45) and update $\mathbf{w}_k^{\iota+1}$ by $\mathbf{w}_k^{\iota}$.
$\iota=\iota+1$.
end for
end for
until convergence of the objective function in (44).
Obtain optimized thresholds Γ_{SR}.
repeat
for $i=1,\ldots,T$ **do**
for $q=1,\ldots,Q$ **do**
Solve (43) and update $\mathbf{w}_q^{\iota+1}$ via $\mathbf{w}_q^{\iota}$.
$\iota=\iota+1$.
end for
end for
until convergence of the objective function in (30).

After iteratively solving (45) until the objective function in (44) converges, the resulting communication BF vectors are merged with the sensing BF vectors to provide the overall

$$\left\|\mathbf{a}^{H}\left(\phi_{m_t}\right)\mathbf{W}_{m_t,q}\right\|^{2}\geq2\mathbb{R}\left\{\left(\mathbf{W}_{m_t,q}^{\iota}\right)^{H}\mathbf{a}\left(\phi_{m_t}\right)\mathbf{a}^{H}\left(\phi_{m_t}\right)\mathbf{W}_{m_t,q}\right\}-\left\|\mathbf{a}^{H}\left(\phi_{m_t}\right)\mathbf{W}_{m_t,q}^{\iota}\right\|^{2}. \tag{38}$$

initial value of (43). The process of IA algorithm is summarized in Algorithm 1.[4]

Additionally, the complexity of IA algorithm is computed as $\mathcal{O}\left(\iota_{\max}TK \max\min\left(\gamma_s\left(\mathbf{w}\right)\right) + \iota'_{\max}TQ\max\left(\gamma_s\left(\mathbf{w}\right)\right)\right)$, where $\iota_{\max}$ and $\iota'_{\max}$ represent the maximum number of iterations in solving (41) and (45), respectively. The algorithmic complexity is mainly due to two terms, i,e, solving for the secure rate and solving for maximizing the target sensing SNR.

C. Joint Sensing and Communication Bf Vectors Optimization

This subsection describes the details of the JSC-BF algorithm. Different from the PA and IA algorithms, which optimize the communication BF and sensing BF separately, the JSC-BF algorithm achieves joint optimization in the ISAC-MSC system.[5] It is noticeable that the aim of IA algorithm provides a tight upper bound for (28). To further improve the accuracy, we design the JSC-BF algorithm with BF vectors. The optimization problem (28) can be transformed into a semidefinite program, which allows applying a semidefinite relaxation for the non-convex objective. The core idea of IA algorithm is changing the problem into a convex solution via relaxation, using a convex solver to obtain an optimal solution, and transferring the solution of the relaxed problem back to the original problem.

For updating the optimization problem as the semi-definitive programming (SDP), the channel matrix and BF matrix are defined as $\mathbf{H}_k$ and $\mathbf{W}_q$, respectively. $[\mathbf{U}_{m_t}]_{nn'} \in \mathbb{R}^{M_tN_t\times M_tN_t}$ is introduced to select the LEO satellite matrix for satisfying the constraints, where each element of the matrix takes the value 0 or 1. To write the target sensing SNR in a compact form, the intermediate value of the solution is defined as

$$\mathbf{E} = \sum_{m_t\in\mathcal{M}_t}\sum_{m_r\in\mathcal{M}_r}\mu\sigma^2_{m_r,m_t}\mathbf{U}_{m_t}\tilde{\mathbf{a}}\tilde{\mathbf{a}}^H\mathbf{U}_{m_t}, \tag{46}$$

where $\tilde{\mathbf{a}} = \left[\mathbf{a}(\phi_1)^T,\ldots,\mathbf{a}(\phi_{m_t})^T,\ldots,\mathbf{a}(\phi_{M_t})^T\right]^T$.

Therefore, the sensing SNR with $\mathbf{E}$ is rewritten as

$$\gamma_s\left(\mathbf{W}\right) = \frac{\mathrm{Tr}\left(\mathbf{E}\sum_{q\in\mathcal{Q}}\mathbf{W}_q\right)}{M_0\sigma^2_{\mathrm{rad}}}. \tag{47}$$

Similarly, the Signal-to-Interference-plus-Noise Ratio (SINR) of user k is calculated to

$$\gamma_k\left(\mathbf{W}\right) = \frac{\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_k\right)}{\sum_{k'\in\mathcal{K}\backslash\{k\}}\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_{k'}\right) + \sum_{j\in\mathcal{J}}\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_j\right) + \sigma^2_k}. \tag{48}$$

[4]The constraint (43b) and (43c) guarantee the existence of an upper bound on the optimization problem, while it can be observed that (12) clearly monotonically increases with the BF vector. Therefore, we obtain that the algorithm is convergent.

[5]It is noteworthy that ISAC-MSC emphasizes the system model, whereas JSC-BF is a joint optimization algorithm targeting BF vectors.

The SINR of eavesdropper j is given as

$$\gamma_j\left(\mathbf{W}\right) = \frac{\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_j\right)}{\sigma^2_j}. \tag{49}$$

Therefore, the constraints are designed as

$$\begin{cases}\left(1+\tilde{\Gamma}_k^{-1}\right)\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_k\right) - \mathrm{Tr}\left(\mathbf{H}_k\sum_{q\in\mathcal{Q}}\mathbf{W}_q\right) \geq \sigma^2_k;\\ \tilde{\Gamma}_j^{-1}\cdot\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_j\right)\geq\sigma^2_j\end{cases} \tag{50}$$

where $\tilde{\Gamma}_k$ and $\tilde{\Gamma}_j$ are the SINR thresholds for users and eavesdroppers, respectively. Let $\tilde{\Gamma}_k + \tilde{\Gamma}_j = \tilde{\Gamma}_{\mathrm{SR}}$, where $\tilde{\Gamma}_{\mathrm{SR}}$ is the value based on modifications of Γ_{SR}.

To obtain the SINR thresholds regarding the users and eavesdroppers, two optimization problems are formulated to solve $\tilde{\Gamma}_k$ and $\tilde{\Gamma}_j$ as

$$\max_{\tilde{\Gamma}_k}\left(1+\tilde{\Gamma}_k^{-1}\right)\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_k\right) - \mathrm{Tr}\left(\mathbf{H}_k\sum_{q\in\mathcal{Q}}\mathbf{W}_q\right) \tag{51a}$$

$$\text{s.t. } \tilde{\Gamma}_k + \tilde{\Gamma}_j = \tilde{\Gamma}_{\mathrm{SR}}, \forall k\in\mathcal{K}, \forall j\in\mathcal{J} \tag{51b}$$

and

$$\max_{\tilde{\Gamma}_j}\tilde{\Gamma}_j^{-1}\cdot\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_j\right) \tag{52a}$$

$$\text{s.t. } \tilde{\Gamma}_k + \tilde{\Gamma}_j = \tilde{\Gamma}_{\mathrm{SR}}, \forall k\in\mathcal{K}, \forall j\in\mathcal{J}. \tag{52b}$$

The coupling problem between $\tilde{\Gamma}_k$ and $\tilde{\Gamma}_j$ is addressed by fixing one to solve the other, and performing alternating iterations. Based on the above calculations, (50) can be rewritten as (53), shown at the bottom of the page.

It is worth noting that $\mathbf{H}_k$ and $\mathbf{W}_q$ are constrained to be convex hermitian positive semi-definiteness matrices. As a result, the optimization problem is formulated in the SDP form at the i-th time slot as

$$\max_{\mathbf{W}_q}\ \mathrm{Tr}\left(\mathbf{E}\sum_{q\in\mathcal{Q}}\mathbf{W}_q\right) \tag{54a}$$

$$\text{s.t. } (54)\ , \sum_{q\in\mathcal{Q}}\mathrm{Tr}\left(\mathbf{U}_{m_t}\mathbf{W}_q\right)\leq P_m, \forall m_t\in\mathcal{M}_t, \tag{54b}$$

$$\mathrm{rank}\left(\mathbf{W}_k\right) = 1, \forall k\in\mathcal{K}, \tag{54c}$$

$$\mathrm{rank}\left(\mathbf{W}_j\right) \leq J, \forall j\in\mathcal{J} \tag{54d}$$

where (54c) and (54d) can be relaxed by removing the rank constraints [38].

The optimization problem (54) is able to be resolved via CVX and convex SDP solvers, and gains $\mathbf{w}_q$ based on Singular Value Decomposition (SVD) about $\mathbf{W}_q$ [39]. The procedure of the JSC-BF algorithm is summarized in Algorithm 2. The complexity of the JSC-BF algorithm is $\mathcal{O}\left(T\left(KN_k + JN_j\right)\alpha\log\left(\frac{1}{o}\right) + TQ^2\right)$, where N_k and N_j represent the number of iterations for solving $\tilde{\Gamma}_k$ and $\tilde{\Gamma}_j$

$$\left(1+\tilde{\Gamma}_k^{-1}\right)\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_k\right) + \tilde{\Gamma}_j^{-1}\cdot\mathrm{Tr}\left(\mathbf{H}_k\mathbf{W}_j\right) - \mathrm{Tr}\left(\mathbf{H}_k\sum_{q\in\mathcal{Q}}\mathbf{W}_q\right) \geq \sigma^2_k + \sigma^2_j. \tag{53}$$

Algorithm 2 JSC-BF

Require: $[\mathbf{U}_{m_t}]_{nn'}$, T, K, J.
repeat
for $i = 1, \ldots, T$ **do**
for $k = 1, \ldots, K$ **do**
Solve (51) and obtain $\tilde{\Gamma}_k$.
end for
for $j = 1, \ldots, J$ **do**
Solve (52) and obtain $\tilde{\Gamma}_j$.
end for
end for
until convergence of the function in (50).
Output optimized thresholds $\tilde{\Gamma}_k$ and $\tilde{\Gamma}_j$.
repeat
for $i = 1, \ldots, T$ **do**
Solve (54) and obtain $\mathbf{W}_q$.
end for
until convergence of the objective function in (30).

iterations, respectively. α and o are the parameter for self-concordant barrier and the precision, respectively [40].

VI. Joint CRB Minimization Strategy

The preceding section effectively enhanced the reliability of object detection by maximizing the perceived SNR. Building upon this foundation, to further meet the demands of high-precision sensing tasks, the current section introduces the Cramér-Rao Bound (CRB) as an extension of the performance metric. The estimation accuracy of the target position $\Lambda = [\dot{x}, \dot{y}, \dot{z}]^T$ is bounded by the CRB, which is the inverse of the system-level Fisher Information Matrix (FIM). For the MSC system, the FIM $\mathbf{J}(\Lambda)$ is the sum of information contributions from all collaborative receiving satellites $m_r \in \mathcal{M}_r$ over T time slots

$$J_{uu'} = \sum_{m_r \in \mathcal{M}_r} \sum_{i=1}^{T} \mathrm{Tr}(\mathbf{C}_{m_r}^{-1}(\Lambda) \frac{\partial \mathbf{C}_{m_r}(\Lambda)}{\partial \Lambda_u} \times \mathbf{C}_{m_r}^{-1}(\Lambda) \frac{\partial \mathbf{C}_{m_r}(\Lambda)}{\partial \Lambda_{u'}}), u, u' \in [\dot{x}, \dot{y}, \dot{z}], \tag{55}$$

where $\mathbf{C}_{m_r}(\Lambda)$ is the covariance matrix of the received echo at satellite m_r. The optimization problem can be formulated as

$$\min_{\Omega_i, \mathbf{w}} \sum_{i \in \mathcal{W}} \mathrm{Tr}\left(\mathbf{J}^{-1}(\Lambda)\right) \tag{56a}$$

$$\text{s.t.} \min_{j \in \mathcal{J}} \left(\sum_{i \in \mathcal{W}} R_k[i] - R_j[i] \right) \geq \Gamma_{\mathrm{SR}}, \forall k \in \mathcal{K}, \tag{56b}$$

$$\|\mathbf{w}_{m_t}\|^2 \leq P_m, \forall m_t \in \mathcal{M}_t, \tag{56c}$$

$$\Omega_i = \left\{ \mathcal{M}_t^k, \mathcal{M}_r^j \right\}, \forall k \in \mathcal{K}, \forall j \in \mathcal{J}, \tag{56d}$$

$$\left| \mathcal{M}_t^k \right| = \left| \mathcal{M}_r^j \right| = M_0, \forall k \in \mathcal{K}, \forall j \in \mathcal{J}. \tag{56e}$$

Problem (56) is a MINCP problem due to the matrix inversion in the objective and the binary constraints on Ω_i.

To handle the non-convex objective, we fix the satellite assignment Ω_i and transform the problem into a SDP. By introducing the BF matrix $\mathbf{W}_q$ of JSC-BF algorithm and an auxiliary variable $\mathbf{X} \in \mathbb{R}^{3\times 3}$, we utilize the Schur complement to convert $\mathbf{X} \geq \mathbf{J}^{-1}(\Lambda)$ into a Linear Matrix Inequality (LMI) as [41]

$$\begin{bmatrix} \mathbf{X} & \mathbf{I}_3 \\ \mathbf{I}_3 & \mathbf{J}(\mathbf{W}_q) \end{bmatrix} \geq 0, \tag{57}$$

where $\mathbf{J}(\mathbf{W}_q)$ is linearized using the inner approximation method at the current iteration. $\mathbf{I}_3$ denotes the 3×3 identity matrix. Therefore, the sub-problem for $\mathbf{W}_q$ at time slot i is rewritten as

$$\min_{\Omega_i, \mathbf{W}_q} \sum_{i \in \mathcal{W}} \mathrm{Tr}\left(\mathbf{X}\right) \tag{58a}$$

$$\text{s.t. } 54(c), \min_{j \in \mathcal{J}} \left(\sum_{i \in \mathcal{W}} R_k\left(\mathbf{W}\right) - R_j\left(\mathbf{W}\right) \right) \geq \Gamma_{\mathrm{SR}}, \quad \forall k \in \mathcal{K}, \tag{58b}$$

$$\mathbf{W}_q \geq 0, \mathrm{rank}(\mathbf{W}_q) = 1, \forall q \in \mathcal{Q}. \tag{58c}$$

Constraint (58c) signifies that $\mathbf{W}_q$ must be a positive semi-definite matrix, and the rank-one constraint ensures that the optimized matrix can be decomposed back into a unique BF vector w_q for signal transmission. The FIM is coupled with the BF matrix $\mathbf{W}_q$ through the derivative $\frac{\partial \mathbf{C}_{m_r}}{\partial \Lambda_s}$. By applying the chain rule to the signal model, the explicit expansion is obtained by

$$\frac{\partial \mathbf{C}_{m_r}(\Lambda)}{\partial \Lambda_u} = \mu \sum_{q \in \mathcal{Q}} \left(\frac{\partial \mathbf{A}}{\partial \Lambda_u} \mathbf{W}_q \mathbf{A}^H + \mathbf{A} \mathbf{W}_q \frac{\partial \mathbf{A}^H}{\partial \Lambda_u} \right) + \frac{\partial \mu}{\partial \Lambda_u} \sum_{q \in \mathcal{Q}} \mathbf{A} \mathbf{W}_q \mathbf{A}^H, \tag{59}$$

where $\mathbf{A} = \sqrt{\sigma_{m_r, m_t}^2} a(\phi_{m_r}) \tilde{a}^H$ is the composite array response matrix, and $\tilde{a}^H$ is the conjugate transpose of the combined array response vector for the satellite constellation launch.

To solve (58) in the CVX solver, we employ successive convex approximation. In the current iteration, the term $\mathbf{C}_{m_r}^{-1}(\Lambda)$ in equation (58) is treated as a constant matrix, whose value is computed from the BF result of the previous iteration. Furthermore, the coupling relationship between satellite allocation Ω_i and BF matrix $\mathbf{W}_q$ is resolved through an alternating optimization method, where Ω_i can be obtained using previously proposed relevant algorithms.

VII. Numerical Results

A. Simulation Settings and Benchmarks

This section provides the numerical results to evaluate the performance of the proposed algorithms in the ISAC-MSC system. The numerical verification results exceed 200, with the location information for each LEO satellite, eavesdroppers, and users being randomly distributed in each simulation. To assess the effectiveness of the proposed algorithms in a realistic setting, the latitude and longitude of ISAC-MSC system elements are located at $[-45°\mathrm{S}, 45°\mathrm{N}]$ and $[45°\mathrm{E}, 90°\mathrm{E}]$. Through the synergy between the proposed BF algorithm and beamwidth, accurate distinction between eavesdroppers and

TABLE II
SIMULATION SETTINGS

Parameter	Value
Frequency, f_c	11.7 GHz
LEO satellites altitude	$500 - 2000$ km
LEO satellite beam width	50 km
Number of transmitter LEO satellites, M_t	200
User k's noise power, σ_k^2	-126 dBW
Number of satellites served per user, M_0	6
Number of users, K	30
Number of eavesdroppers, J	30
Eavesdroppers noise power, σ_j^2	-97.2 dBW
Number of transmitting antennas via BF, N_t	4
Number of antennas for receiver LEO satellite, N_r	4
User receive antenna gain, G_r	35.7 dBi
LEO satellite transmit antenna gain, G_t	64.9 dBi
Number of time slots, T	16
Power budget per LEO satellite, P_m	25 dBW
Number of receiver LEO satellites, M_r	200
Radar receiver noise power, σ_{rad}^2	-126 dBW
Variance of bi-static RCS, σ_{m_r,m_t}^2	0.1
LEO satellites travel speed, v_0	7.9 km/s

users can be achieved. Subsequently, simulations are conducted to evaluate performance metrics such as target sensing SNR and secure communication rate, to assess the system's overall performance. The LEO satellite-related parameters and collaboration methods refer to the 3rd Generation Partnership Project (3GPP) Technical Report (TR) 38.811 and TR 38.821 [42], [43]. Unless otherwise specified, the definitions of these parameters are summarized in TABLE II.

We verify the excellence of proposed algorithms by selecting several different comparison benchmarks as follows.

1) SHP-PA [44]: the SHP algorithm is considered as a comparison baseline for optimizing LEO satellite assignment. Since PA is a modification of the traditional algorithm, it serves as a benchmark with the BF vectors. The SHP algorithm and PA optimize the corresponding variables and perform alternating iterations.
2) SHP-IA: the SHP algorithm still optimizes the LEO allocation by fixing $\mathbf{w}_q$. Then, IA optimizes the BF vectors and combines both algorithms.
3) SHP-JSC-BF: the SHP is introduced as the benchmark and combined with JSC-BF.
4) DP-PA, DP-IA, DP-JSC-BF: the DP algorithm is used to optimize the LEO allocation and is combined with PA, IA, and JSC-BF with optimized BF vectors, respectively, and finally iterates the result of (13).
5) CP-PA, CP-IA, CP-JSC-BF: similar to the above, we combine the CP algorithm with PA, IA, and JSC-BF to analyze the sensing and communication performance.

B. Sensing Performance

Fig. 3 plots the target sensing SNR versus the power budget of the LEO satellite with multiple comparison algorithms. The proposed algorithms all demonstrate significantly superior performance to the corresponding SHP benchmark algorithms.

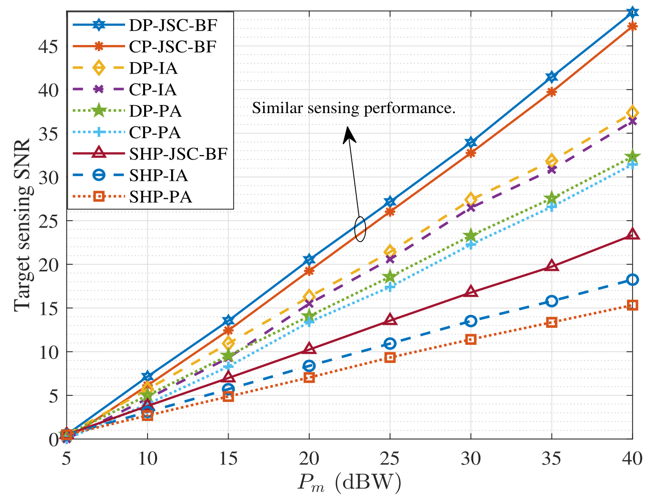

Fig. 3. Target sensing SNR versus power budget of LEO satellite in the ISAC-MSC system.

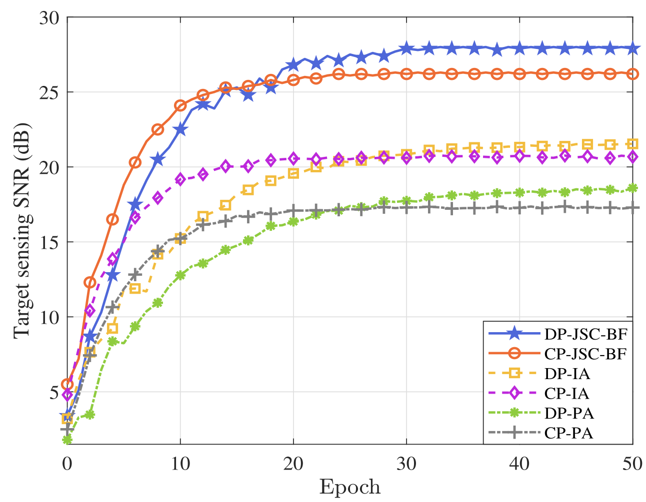

Fig. 4. The convergence of the proposed algorithms.

It can be observed that the sensing performance of CP and DP algorithms are similar and superior. This is because ξ_{m_tk} in (20) is a binary variable, and the DP algorithm is more suitable for the LEO satellite allocation problem. However, ξ_{m_tk} is relaxed, generating a minor performance degradation in the CP algorithm. Another phenomenon is that the JSC-BF algorithm has significant differences compared to the PA and IA algorithms. The PA algorithm is optimized for sensing power, which creates a gap with the results obtained by directly solving for the BF vectors. In the meantime, the IA algorithm is an approximate result for (13a), leading to a degradation of sensing performance. It is evident that the DP-JSC-BF algorithm is more suitable for sensing eavesdroppers with acceptable complexity.

Fig. 4 illustrates the convergence performance of the proposed CP-based and DP-based algorithms. It can be observed that as the number of epochs increases, the target sensing SNR of all algorithms gradually improves and eventually reaches a stable state, which verifies the convergence of the proposed alternating iterative framework. Specifically, the DP-JSC-BF

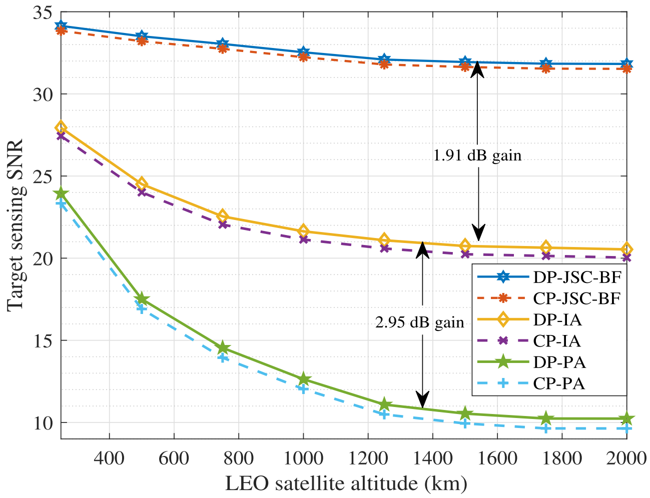

Fig. 5. Target sensing SNR versus altitude of LEO satellite with $P_m = 25\,\text{dBW}$ in the ISAC-MSC system.

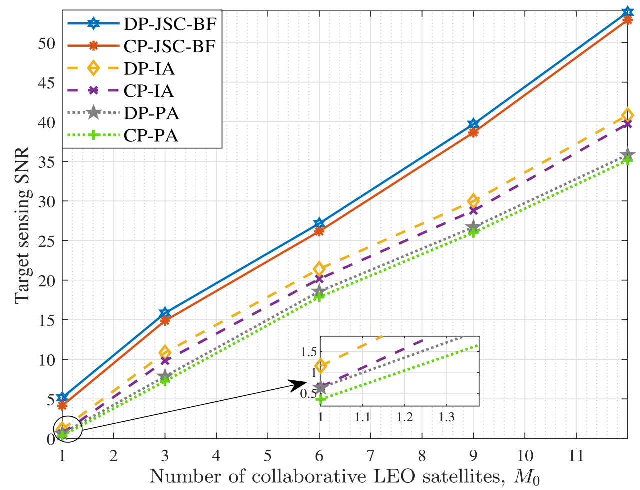

Fig. 6. Target sensing SNR versus the LEO satellite collaborative number.

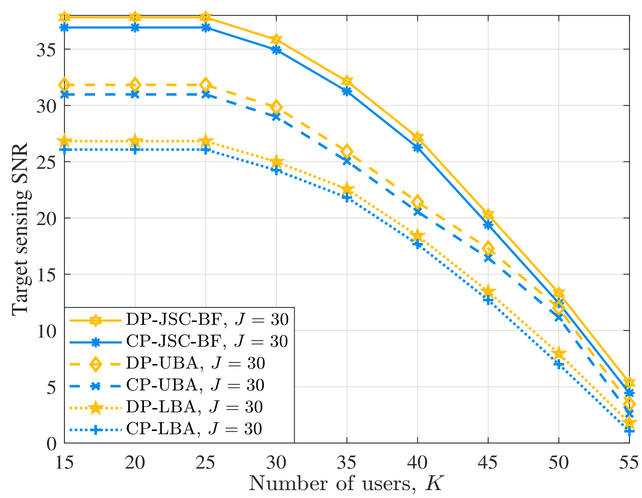

Fig. 7. Target sensing SNR versus the number of users at $J = 30$.

and CP-JSC-BF algorithms achieve the highest converged SNR compared to the IA and PA benchmarks. Notably, the CP-based algorithms exhibit a faster initial rising speed in the early iterations, which is attributed to the continuous relaxation that allows for a more fluid search in the solution space. In contrast, the DP-based algorithms, while requiring more iterations to converge, eventually yield a slightly higher sensing SNR because the discrete mapping is more consistent with the binary nature of the satellite assignment matrix $\xi_{m_t k}$. This further confirms that the DP-JSC-BF algorithm provides a superior balance between sensing precision and global optimality within the ISAC-MSC system.

The design of the LEO satellite orbital altitude affects the sensing capability in the case of a fixed beam width. The LEO satellite altitude cannot be changed, but the impact on sensing performance of their altitude deployment needs to be analyzed, as shown in Fig. 5. We observe that the JSC-BF based algorithm outperforms the IA algorithm by 1.91 dB, while the IA algorithm has a gain of 2.95 dB compared to the PA algorithm. This illustrates that the JSC-BF algorithm has an advantage in distinguishing between users and eavesdroppers. As the LEO satellite's orbital altitude increases, the angle between the LEO satellite observing users and eavesdroppers becomes smaller at constant beam width, which is the reason for the target sensing SNR to stabilize. Due to the high-speed movement of LEO satellites resulting in rapid changes in the angle between LEO satellites and ground targets, the selection of the satellite altitude for the ISAC-MSC system is crucial.

Compared to traditional single-satellite scenarios, the MSC system is able to enhance the ISAC signal transmission in different orbits and positions. Specifically analyzed, Fig. 6 illustrates the relationship between target sensing SNR and the LEO satellite collaborative number. It is worth noting that the single satellite scenario is highlighted at $M_0 = 1$, where a single satellite has limitations for the sensing enhancement. Multi-satellite distributed collaboration delivers better spatial information resolution than single satellites. It uses cooperative BF for spatial signal energy reshaping, enhancing legitimate channel capacity, suppressing eavesdropping, and greatly improving secure communication capacity. As the number of collaborating satellites increases, the sensing performance grows approximately linearly in the ISAC-MSC system, which reflects that the algorithms excellently achieve the optimization objective based on (12). It follows that the ISAC-MSC system is highly scalable, which means that the modes of collaboration among LEO satellites can be changed flexibly.

As a further advance, Fig. 7 plots the target sensing SNR versus the number of users at $J = 30$. One phenomenon is that sensing performance is steadily at a high level when the number of users is less than the number of eavesdroppers, which is because MSC systems can easily be distinguished from them. However, the satellite beam covers a large area, and an increase in user number leads to an expansion of the ground elements, thus worsening the sensing performance of the system. Another phenomenon is that compared to the IA algorithm, the JSC-BF algorithm improves 0.76 dB and 0.79 dB for the DP and CP schemes at $K = 25$, respectively. Simultaneously, the DP-IA and CP-IA algorithms have gains of 0.79 dB and 0.89 dB compared to the DP-PA and CP-PA

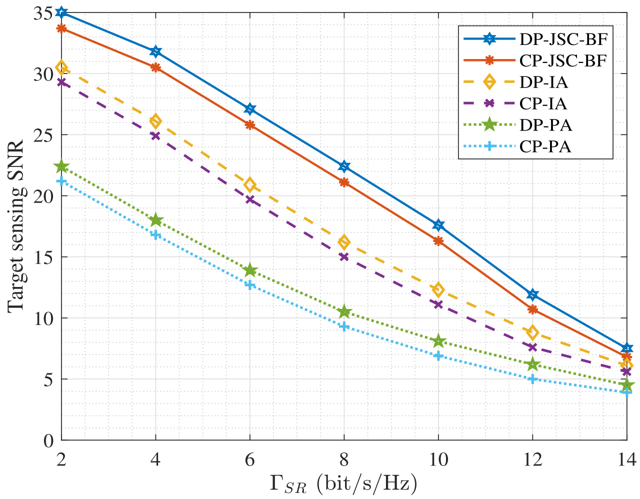

Fig. 8. Target sensing SNR versus Γ_{SR} at $P_m = 25\mathrm{dBW}$.

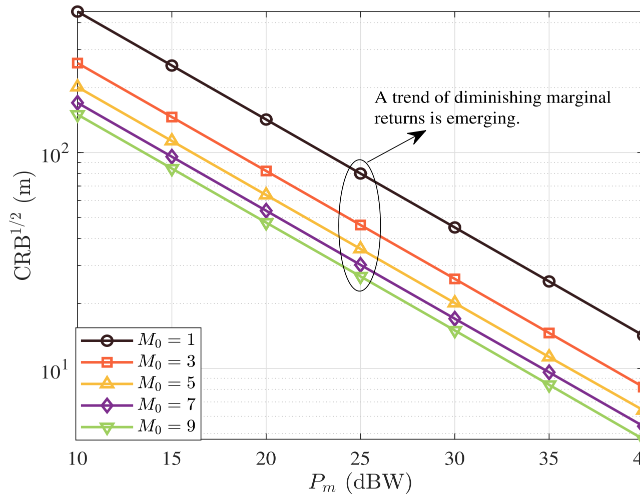

Fig. 9. Position estimation error bound CRB$^{1/2}$ versus P_m for varying collaboration scales M_0.

algorithms for $K = 25$. Furthermore, the design of ISAC-MSC system can be further optimized according to the distribution of users and eavesdroppers on the ground.

Fig. 8 draws the relationship between the secure communication rate threshold Γ_{SR} and the target sensing SNR. According to optimization problem (13), the ISAC-MSC system needs to maximize γ_s while satisfying the minimum secure communication rate constraint, which constitutes a resource balancing problem. It can be observed that Γ_{SR} exhibits a negative correlation with the target detection SNR. This is because the marginal effects of resource allocation differ across different threshold ranges. At lower thresholds, the security constraint is less stringent, allowing sensing performance to remain at a relatively high level with a gradual decline. When the threshold increases to a certain point, the security constraint becomes the primary bottleneck, necessitating a drastic reallocation of resources that causes sensing performance to drop rapidly. At very high thresholds, nearly all system resources are allocated to ensuring communication, causing sensing performance to decrease to a very low level and stabilize.

While maximizing the sensing SNR ensures that the system can detect the presence of a target, the CRB is introduced to characterize the precision of the sensing task. The necessity of the CRB analysis lies in its ability to map signal resource allocation directly to physical localization accuracy. As depicted in Fig. 9, the CRB$^{1/2}$ quantifies the minimum achievable standard deviation of the target's spatial coordinates. This metric is vital for security applications, as it determines whether the ISAC-MSC system can accurately track an eavesdropper or merely acknowledge its existence. The numerical results show that with a sufficient power budget and multi-satellite collaboration, the CRB$^{1/2}$ can reach the low-meter scale, e.g., approximately 10^1 m and below. Increasing the number of collaborative satellites M_0 from 1 to 9 significantly compresses the error bound. This improvement stems from the enhanced spatial diversity and more favorable observation geometries, which mitigate the geometric dilution of precision inherent in single-satellite sensing. A trend of diminishing marginal returns is emerging as M_0 increases. While higher M_0 continues to lower the error bound, the rate of improvement slows down after $M_0 = 5$, suggesting that the system enters a regime of high-precision stability. This provides a critical design guideline for balancing sensing resolution against the synchronization and communication overhead required for large-scale multi-satellite coordination.

It is worth noting that the simulation design for target sensing SNR and CRB in this paper is interrelated. For instance, the enhancement of perceived SNR in Fig. 3 provides the fundamental signal energy for target detection, while CRB, such as in Fig. 9, further quantifies the accuracy of target localization based on this foundation. Simulation results show that as the transmit power budget P_m increases, the upward trend in target perception SNR aligns with the downward trend in position estimation error bound CRB$^{1/2}$, indicating that higher signal strength contributes to improved parameter estimation accuracy. In addition, despite the sensing SNR increasing approximately linearly with M_0, the CRB exhibits a pronounced diminishing marginal benefit beyond $M_0 \approx 5$. This phenomenon indicates the existence of a trade-off point between system performance and overhead at $M_0 \approx 5$. At this point, the system has reached a stable zone for high-precision positioning. Continuously adding collaborative nodes will significantly increase system complexity and overhead, without yielding significant improvements in positioning accuracy.

C. Secrecy Performance

From the viewpoint of security communications, we analyze the secrecy rate of users. Thus, the following simulation process is to minimize the communication secrecy rate, i.e., $\sum_{k\in\mathcal{K}} \left[\min_{j\in\mathcal{J}} (R_k - R_j)\right]^+$. The simulation parameters still remain the same as in Section VI-A.

Fig. 10 plots the minimum communication secrecy rate versus the power budget, which responds to the security of the ISAC-MSC system in the worst state. It can be observed that the communication security performance grows almost linearly

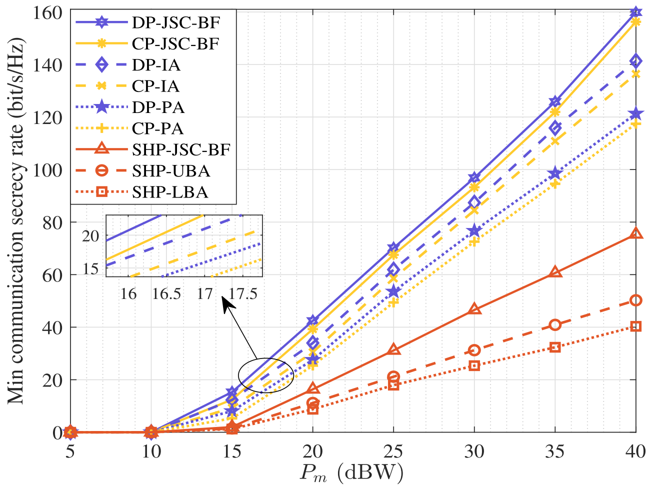

Fig. 10. The minimum communication secrecy rate versus the LEO satellite power budget.

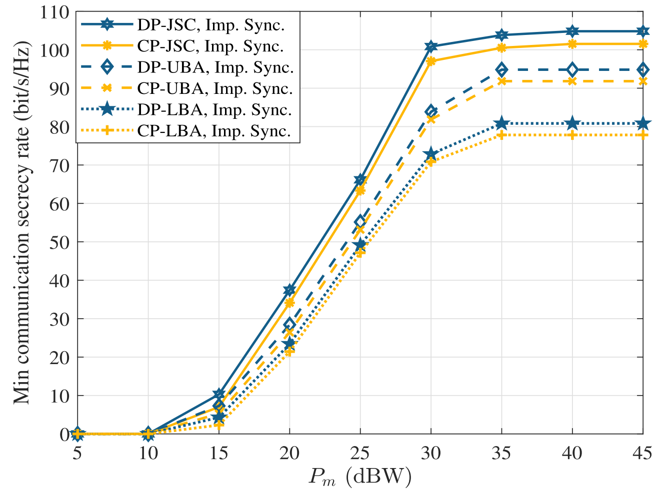

Fig. 12. The minimum communication secrecy rate versus P_m under imperfect MSC synchronization.

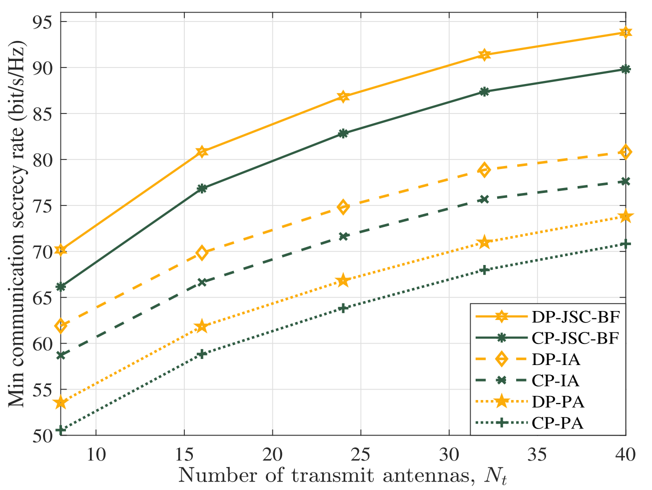

Fig. 11. The minimum communication secrecy rate versus number of transmit antennas at $P_m = 25\,\text{dBW}$.

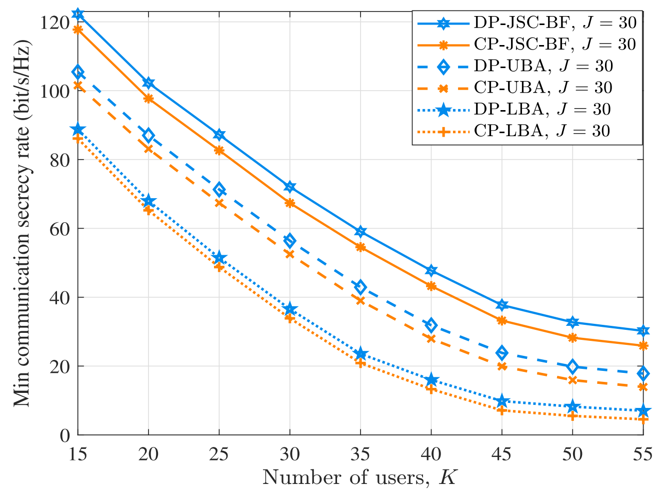

Fig. 13. The minimum communication secrecy rate versus the number of users at $J = 30$.

for P_m more than 15 dBW. This is because the MSC is able to ensure hardware stability and strong anti-jamming capability over a wide range of power. Compared to SHP algorithms, CP and DP-based algorithms also have excellent performance in terms of communication secrecy rate. Moreover, the JSC-BF algorithms still exist with the 0.6 dB to 0.65 dB boosting over the IA algorithms in $P_m = 25\,\text{dBW}$. Also, the IA algorithms have a gain of 0.6 dB to 0.71 dB for the PA algorithms. The power of each LEO satellite is limited by the payload capacity, which affects the secrecy performance of the ISAC-MSC system.

From the viewpoint of security transmission, the number of LEO satellite transmitter antennas can directly affect the accuracy of the BF. Fig. 11 shows the minimum communication secrecy rate versus the number of transmit antennas. As the number of antennas increases, the communication security performance also rises, but its improvement becomes slower. Due to the fact that the number of eavesdroppers and users remains constant, simply adding the number of antennas does not result in a linear increase. More antennas provide higher gain, allowing the energy of the beam to maintain its strength over greater distances, yet improved coverage can make it easier for eavesdroppers to attack users. It can be clearly found that DP-JSC-BF and CP-JSC-BF algorithms have a remarkable minimum communication secrecy rate.

To assess the practical feasibility of the ISAC-MSC system, the impact of imperfect MSC synchronization is investigated. Unlike single satellite systems, collaborative BF among dispersed LEO satellites requires precise carrier phase alignment, which is challenging due to high speed mobility and independent local oscillators. The impairment is modeled by introducing a random phase error term to the channel vector. Let $h_{m_t,k}$ denote the ideal channel state information. The actual equivalent channel with synchronization errors, denoted as $\tilde{h}_{m_t,k} = h_{m_t,k} \cdot e^{j\Delta_{m_t}}$, where Δ_{m_t} represents the residual phase synchronization offset of the m_t-th satellite, following a truncated Gaussian distribution with zero mean and variance σ_Δ^2, i.e., $\Delta_{m_t} \sim \mathcal{N}(0, \sigma_\Delta^2)$.

Building upon the established error model, Fig. 12 illustrates the minimum communication secrecy rate versus P_m,

$$\begin{aligned}\log\left(1+\frac{|\mathbf{g}|^2}{|\mathbf{m}|^2+\sigma^2}\right) &\geq \log\left(1+\frac{|\mathbf{g}^{\iota}|^2}{|\mathbf{m}^{\iota}|^2+\sigma^2}\right)+\frac{|\mathbf{g}|^2}{|\mathbf{m}^{\iota}|^2+\sigma^2}-\frac{|\mathbf{g}^{\iota}|^2}{|\mathbf{m}^{\iota}|^2+\sigma^2}\\ &\geq \log\left(1+\frac{|\mathbf{g}^{\iota}|^2}{|\mathbf{m}^{\iota}|^2+\sigma^2}\right)+\frac{2\mathbb{R}\left\{\left(\mathbf{g}^{\iota}\right)^H\mathbf{g}\right\}}{|\mathbf{m}^{\iota}|^2+\sigma^2}-\frac{|\,\mathbf{g}^{\iota}\|^2}{|\mathbf{m}^{\iota}|^2+\sigma^2}\\ &\quad -\frac{|\mathbf{g}^{\iota}|^2\sigma^2}{\left(|\mathbf{m}^{\iota}|^2+\sigma^2\right)\left(|\mathbf{g}^{\iota}|^2+|\mathbf{m}^{\iota}|^2+\sigma^2\right)}-\frac{|\mathbf{g}^{\iota}|^2\left(|\mathbf{g}|^2+|\mathbf{m}|^2\right)}{\left(|\mathbf{m}^{\iota}|^2+\sigma^2\right)\left(|\mathbf{g}^{t}|^2+|\mathbf{m}^{\iota}|^2+\sigma^2\right)}\\ &= \log\left(1+\frac{|\mathbf{g}^{\iota}|^2}{|\mathbf{m}^{\iota}|^2+\sigma^2}\right)+\frac{2\mathbb{R}\left\{\left(\mathbf{g}^{\iota}\right)^H\mathbf{g}\right\}}{|\mathbf{m}^{\iota}|^2+\sigma^2}-\frac{|\mathbf{g}^{\iota}|^2+\sigma^2}{|\mathbf{m}^{\iota}|^2+\sigma^2}\\ &\quad +\frac{\sigma^2}{\left(|\mathbf{m}^{\iota}|^2+\sigma^2\right)\left(|\mathbf{g}^{\iota}|^2+|\mathbf{m}^{\iota}|^2+\sigma^2\right)}-\frac{|\mathbf{g}|^2+|\mathbf{m}|^2)}{\left(|\mathbf{m}^{\iota}|^2+\sigma^2\right)\left(|\mathbf{g}^{\iota}|^2+|\mathbf{m}^{\iota}|^2+\sigma^2\right)}.\end{aligned} \tag{61}$$

explicitly accounting for the impact of imperfect MSC synchronization. Observing the results, it is evident that while phase synchronization errors inevitably introduce a performance penalty, the secrecy rate maintains a monotonic upward trend as the power budget expands. Notably, the proposed DP-JSC-BF and CP-JSC-BF algorithms demonstrate exceptional robustness, consistently outperforming the IA and PA schemes even in the presence of phase uncertainty. This resilience is attributed to the joint optimization strategy's ability to dynamically adjust BF vectors, thereby effectively compensating for phase mismatches and suppressing information leakage to eavesdroppers in realistic, non-ideal distributed environments.

Fig. 13 plots the minimum communication secrecy rate versus the number of users at $J = 30$. It can be observed that as the number of users becomes larger, the communication security gradually decreases. This indicates that more users are vulnerable to eavesdroppers. Servicing more users within a fixed power budget leads to reduced security. In addition, an enlarged number of users makes it more difficult to sense eavesdroppers, which further complicates distinguishing between user and eavesdropper states. It is clear that the minimum communication secrecy rate is higher for a similar number of users and eavesdroppers in the JSC-BF algorithms. Specifically, based JSC-BF algorithms showed 0.81 dB and 2.76 dB improvement, respectively, in optimal performance compared to the IA and PA algorithms.

VIII. Conclusion

In this paper, we have presented a novel ISAC-MSC security system to maximize target sensing SNR with limited power and a number of collaborative LEO satellites. Firstly, we have decomposed the optimization problem into two subproblems by jointly optimizing LEO satellite assignments and BF vectors, and performing alternative iterations. Then, we have developed the improved CP and DP algorithms to address the optimization problem in the LEO satellite assignment. To optimize BF vectors, we introduce three algorithms: PA, IA and JSC-BF. Additionally, we have discussed the computational complexity of these algorithms. The numerical results have shown that the JSC-BF algorithms have excellent performance for sensing targets and security communication compared to IA and PA algorithms. In contrast to a single satellite system, the proposed ISAC-MSC system generates an excellent performance with approximately linear growth.

Our research is expected to contribute positively to the development of security satellite communications to address new challenges. Furthermore, future ISAC-MSC systems can flexibly switch collaboration methods according to the locations of the user and eavesdropper in time, promoting a more efficient use of resources on the satellites. The ISAC-MSC system can be further applied to complex scenarios such as smart cities, agriculture, and emergency response. Our work facilitates the establishment of a holistic and green ISAC-MSC system and provides a sound theoretical basis for future innovations in satellite communication technology.

Appendix A

The following inequality holds true for any $\mathbf{g}, \mathbf{g}^{\iota} \in \mathbb{C}^{N_1}$ owing to the convexity of $\|\mathbf{g}\|^2$ [37]

$$\|\mathbf{g}\|^2 \geq 2\mathbb{R}\left\{\left(\mathbf{g}^{\iota}\right)^H\mathbf{g}\right\} - \|\mathbf{g}^{\iota}\|^2. \tag{60}$$

For any $(\mathbf{g}, \mathbf{m})$ and $(\mathbf{g}^{\iota}, \mathbf{m}^{\iota}) \in \mathbb{C}^{N_1} \times \mathbb{C}^{N_2}$, there exists inequality (56). By applying the inequality (56) for $|\mathbf{g}|^2 = \left|\mathbf{h}_k^H\mathbf{w}_k\right|^2$ and $|\mathbf{m}|^2 = \sum_{\bar{j}\in\{\mathcal{K},\mathcal{J}\}\setminus\{k\}}\left|\mathbf{h}_k^H\mathbf{w}_{\bar{j}}\right|^2 = \sum_{k'\in\mathcal{K}\setminus\{k\}}\left|\mathbf{h}_k^H\mathbf{w}_{k'}\right|^2 + \sum_{j\in\mathcal{J}}\left|\mathbf{h}_k^H\mathbf{w}_j\right|^2$, a lower bounding concave approximation of $R_k^{\iota}(\mathbf{w})$ is given by (39).

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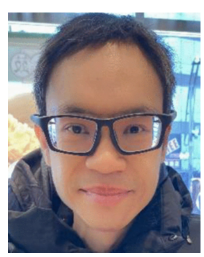

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